



LightGlue [5], and detector-free methods such as LoFTR [6]—has demonstrated substantial improvements in matching robustness under challenging conditions. However, integrating these methods into ISIS-based workflows requires complex middleware, limiting their adoption in planetary photogrammetry. In order to address this barrier, a unified framework is required that provides native Python access to ISIS geometric primitives while supporting integration with deep learning libraries.

**3) Adaptive method selection:** No single matching algorithm performs optimally across the full range of planetary imaging conditions. Classical methods such as SIFT [7] excel on texture-rich imagery with similar illumination, while deep learning methods demonstrate superior performance under large illumination changes or sparse texture. Manual selection of matching strategies requires expert knowledge and is impractical for large-scale processing campaigns. Recent benchmarking on 496 WorldView-3 stereo pairs with LiDAR ground truth [8] provides strong empirical validation for multi-method approaches: no single method succeeded universally across diverse conditions, motivating cascade fallback designs.

The main contributions of our work are as follows:

**1) PyISIS Framework:** We present a comprehensive pybind11 binding framework exposing over 200 ISIS 9.0.0 C++ classes and types to Python, covering camera models, SPICE navigation, control networks, bundle adjustment, projections, photometry, and shape models. The framework enables seamless integration of ISIS photogrammetric capabilities within Python scientific computing workflows.

**2) Unified Matching Pipeline:** We develop an integrated image matching pipeline that unifies classical (SIFT-BF, SIFT-FLANN) and deep learning (SuperGlue, LightGlue, LoFTR) matchers within a tile-based processing framework. The pipeline supports 17 configurable presets spanning different feature extractors and matching strategies, with GPU acceleration for deep learning methods.

**3) Adaptive Routing System:** We propose an adaptive matcher routing system that automatically selects the optimal matching strategy based on quantitative image analysis. The system computes (a) texture sparseness via tile-level SIFT density, gradient magnitude, and GLCM contrast, and (b) lighting differences from SPICE-derived solar elevation and azimuth geometry. A cascade fallback mechanism ensures robust matching by progressively escalating from classical to deep learning methods when quality thresholds are not met. To our knowledge, PyISIS is the first framework to leverage mission-specific ephemeris metadata for illumination-aware matching strategy selection.

**4) Automated Control Network Pipeline:** We present an end-to-end control network construction workflow that integrates PyISIS geometry, adaptive matching, and coordinate transformation, demonstrated on LRO NAC imagery with over 121,000 automatically generated control points.

The remainder of this paper is organized as follows. Section II reviews related work in planetary photogrammetry software, image matching methods, and adaptive routing strategies. Section III describes the PyISIS framework architecture, unified matching pipeline, adaptive routing system, and control

network construction workflow. Section IV presents experimental results on LRO NAC imagery. Section V discusses limitations and future directions, and Section VI concludes the paper.

## II. RELATED WORK

### A. Planetary Photogrammetry Software

The ISIS software suite, maintained by the USGS Astrogeology Science Center, provides the most comprehensive set of planetary image processing tools available [3]. ISIS supports mission-specific camera models for framing, push-broom, and push-frame sensors, with rigorous SPICE-based navigation data integration. The *jigsaw* application implements bundle adjustment for simultaneous refinement of camera pointing and control point coordinates [9]. However, ISIS operates primarily through command-line applications that process ISIS Cube files, limiting its integration with modern Python-based research workflows. Based on this observation, several attempts have been made to provide Python access to ISIS functionality.

AutoCNet, developed by USGS Astrogeology, is a Python library for automated sparse control network generation [10]. It employs computer vision techniques for n-image correspondence identification and integrates with ISIS workflows through intermediate file I/O. While AutoCNet addresses automation needs, it relies on subprocess calls to ISIS applications rather than direct API access, resulting in suboptimal performance for large-scale processing. The Ames Stereo Pipeline (ASP) provides complementary stereo processing capabilities, including semi-global matching and bundle adjustment, with Python bindings through its command-line interface [11]. ASP and ISIS are often used together in planetary mapping workflows.

Recent efforts to provide Python access to ISIS functionality include SpiceQL, a REST/Python/C++ API for SPICE kernel queries [12], and preliminary discussions within the ISIS community about Python bindings via pybind11 [13]. PLIO (Planetary I/O Module) provides Python read/write access for ISIS Cube files and control network data but operates at the file I/O level only, without exposing camera models or computational APIs. However, no prior work has provided comprehensive Python bindings covering the full breadth of ISIS functionality.

Recent work on LRO NAC control network construction includes Wang et al. [14], who presented a method for refining tie points by matching with shaded LOLA DEM data, published in this journal. Their approach uses orthophoto-based matching with DEM-shaded renders as intermediate references to improve tie point accuracy in challenging illumination conditions. While their work addresses the matching refinement stage using terrain-shaded renders, our approach focuses on the initial matching stage with adaptive routing and cascade fallback. The two methods are complementary: DEM-shaded refinement could serve as a post-processing step following our adaptive matching pipeline to further improve tie point quality.

### B. Image Matching Methods for Planetary Imagery

Classical feature-based matching methods, particularly SIFT [7] combined with nearest-neighbor ratio filtering [15] and brute-force or FLANN-based descriptor matching [16], have been the workhorse of planetary photogrammetry for over a decade. These methods perform well on texture-rich imagery with moderate illumination differences but degrade significantly on sparse-texture surfaces (e.g., smooth plains, polar regions) and under large solar geometry variations common in multi-temporal planetary datasets. Comprehensive benchmarking by Ye and Zhou [17] evaluated 13 feature detectors and 12 descriptors across Moon and Mars multisource images, finding that SIFT-based pipelines degrade significantly on sparse-texture surfaces and under large illumination changes.

Deep learning-based image matching has undergone rapid development in recent years. SuperGlue [4] introduced a graph neural network (GNN) architecture with optimal transport for learned keypoint matching, demonstrating improved robustness under viewpoint and illumination changes. LightGlue [5] advanced this paradigm with adaptive early-termination mechanisms that reduce computation on easy image pairs while maintaining accuracy on challenging ones. Both methods operate on detected features, typically from SuperPoint [18] or other learned detectors.

Detector-free methods represent an alternative paradigm. LoFTR [6] employs self-attention and cross-attention transformer layers to establish dense correspondences without explicit feature detection, showing particular strength on textureless scenes. EfficientLoFTR [19] further improves computational efficiency through sparse attention mechanisms, achieving  $2.5\times$  speedup over LoFTR while maintaining accuracy. Dense matching methods such as RoMa [20] and DKM [21] produce pixel-dense correspondences with  $>95\%$  inlier ratios on satellite imagery [8].

In the planetary domain, recent work has evaluated deep learning matchers for specific applications. Geo-LoFTR [22] incorporates geometric context from digital terrain models for Mars rotorcraft localization under challenging illumination. Feature matching evaluations on multisource planetary remote sensing imagery [17] have demonstrated that deep learning methods can outperform classical approaches under challenging conditions, but no single method dominates across all scenarios. Recent benchmarking on 496 WorldView-3 stereo pairs with LiDAR ground truth [8] provides strong empirical validation for multi-method approaches: SIFT+LightGlue achieved best DSM accuracy when matches existed, DKM achieved  $>95\%$  inlier ratios and best spatial coverage, and SIFT alone had the lowest success rate under seasonal transitions. No single method succeeded universally, motivating our cascade fallback design.

### C. Adaptive Method Selection

The concept of adaptive feature selection and matching strategy has gained traction in visual SLAM and robotics. CHAMELEON-SLAM [23] employs a lightweight scene classifier to switch between XFeat, ALIKED, and SuperPoint feature extractors based on scene characteristics. Light-SLAM

[24] integrates LightGlue into a visual SLAM system specifically designed for challenging lighting conditions. Hybrid approaches combining classical and learned features [25] have demonstrated improved robustness in mixed environments.

In remote sensing, adaptive matching networks [26] have been proposed for multimodal satellite image registration, and weak-texture matching methods [27] address the challenge of sparse features in certain terrain types. AMES (Adaptive Matching with Enhanced Sketches) combines edge-based sketch features with adaptive strategies for multi-modal matching without manual filter tuning.

The existing landscape reveals three critical gaps that our work addresses:

**Gap 1: No comprehensive Python bindings for ISIS.** Existing Python access is limited to subprocess wrappers (AutoCNet [10], Pysis, Kalasiris), file-level I/O without computational API access (PLIO), or narrow-scope APIs for specific subsystems (SpiceQL [12]). No prior framework provides simultaneous pybind11-based access to ISIS camera models, SPICE navigation, control networks, bundle adjustment, projections, photometry, and shape models.

**Gap 2: No adaptive routing for planetary photogrammetry.** While adaptive method selection has been explored in terrestrial SLAM (CHAMELEON-SLAM [23], Light-SLAM [24]) and remote sensing (adaptive matching networks [26]), these approaches either use learned scene classifiers requiring terrestrial training data or analyze generic image features without leveraging mission-specific metadata. Table I compares existing adaptive methods against our approach. No prior work exploits SPICE-derived solar geometry—physically precise measurements of solar elevation and azimuth—for illumination-aware matching strategy selection in planetary contexts.

TABLE I  
COMPARISON OF ADAPTIVE MATCHING SYSTEMS

| System               | Domain           | Routing<br>Signal | Train<br>Data | Planet<br>Meta |
|----------------------|------------------|-------------------|---------------|----------------|
| CHAMELEON [23]       | SLAM             | Learned           | Yes           | No             |
| Light-SLAM [24]      | SLAM             | Fixed             | Yes           | No             |
| Adaptive Net [26]    | RS               | Features          | Yes           | No             |
| <b>PyISIS (ours)</b> | <b>Planetary</b> | <b>SPICE</b>      | <b>No</b>     | <b>Yes</b>     |

Our approach differs fundamentally in three respects: (1) it requires no labeled training data, addressing the absence of planetary matching datasets; (2) it exploits physically precise solar geometry measurements rather than learned approximations; and (3) it is domain-specific to airless-body illumination extremes (absent atmospheric scattering, permanent shadow regions) that terrestrial classifiers cannot capture.

**Gap 3: No integrated pipeline bridging ISIS geometry with deep learning matching.** While individual deep learning matchers have been evaluated on planetary imagery [17], [28], no framework combines (a) rigorous ISIS camera model geometry, (b) multiple deep learning matchers with cascade fallback, and (c) automated control network construction with coordinate transformation from projected DOM space back to original image coordinates.

Our work addresses all three gaps through the PyISIS framework, the unified matching pipeline with adaptive routing, and the end-to-end control network construction workflow.

### III. METHODOLOGY

This section describes the complete methodology for planetary photogrammetric processing using the PyISIS framework. The methodology consists of four main components: (A) the PyISIS framework architecture providing Python bindings for ISIS 9.0.0, (B) the unified image matching pipeline supporting classical and deep learning methods, (C) the adaptive routing system with SPICE-derived illumination analysis, and (D) the control network construction workflow.

#### A. PyISIS Framework Architecture

PyISIS follows a “thin binding” philosophy that exposes the ISIS C++ API with minimal abstraction overhead, preserving the semantics and performance characteristics of the underlying library while providing Pythonic ergonomics. The framework is built on pybind11 [29], chosen for its efficient type conversion, support for C++17 features, and compatibility with the scientific Python ecosystem. According to the characteristics of planetary photogrammetric processing, the framework is organized into 37 C++ binding source files, each responsible for a coherent set of ISIS classes.

**Core I/O Module** (Cube, Brick, LineManager, TileManager, BandManager): Provides both low-level buffer access and high-level read/write operations for ISIS Cube files. The binding supports multi-band imagery with arbitrary pixel types and preserves ISIS-specific metadata through PVL (Parameter Value Language) label access. As can be seen in typical planetary processing workflows, efficient I/O is critical when processing large-scale datasets where individual LRO NAC images exceed  $5000 \times 50,000$  pixels (8 GB per cube).

**Camera Model Module** (50+ mission cameras): Binds the complete ISIS camera model hierarchy, including FramingCamera, LineScanCamera, PushFrameCamera, RollingShutterCamera, and CSMCamera. Mission-specific implementations cover LRO NAC/WAC, MRO HiRISE/CTX, Kaguya TC/MI, TGO CaSSIS, Hayabusa2 ONC, OSIRIS-REx OCAMS, Apollo Metric/Panoramic, Viking Orbiter, and Voyager ISS/NA. Each camera model exposes SetImage(), GroundPoint(), SpacecraftPoint(), SunPosition(), and pixel resolution methods, enabling rigorous geometric computations directly from Python. Based on this idea, researchers can perform forward and inverse projections without invoking ISIS command-line applications through subprocess calls.

**SPICE Navigation Module** (Spice, SpicePosition, SpiceRotation, Kernels): Exposes ALE-derived SPICE kernel loading and querying, including spacecraft ephemeris, instrument pointing, body-fixed frames, and solar geometry computation. The SPICE module is critical for the adaptive routing system’s illumination analysis, providing access to solar elevation and azimuth angles with arc-second precision.

**Control Network Module** (ControlNet, ControlPoint, ControlMeasure, BundleControlPoint, BundleObservation, BundleSettings, BundleResults, BundleSolutionInfo): Provides full programmatic access to ISIS control network creation, editing, and bundle adjustment configuration. This module enables the construction of control networks directly from Python without requiring intermediate file I/O through ISIS’s pointreg or autoseed programs.

**Projection Module** (16 projection types): Binds all major map projections used in planetary cartography, including Equirectangular, SimpleCylindrical, Mercator, PolarStereographic, LambertConformal, TransverseMercator, and Orthographic, with both planetocentric and planetographic coordinate support.

**Photometry Module** (Hapke, Lambert, Minnaert, HapkeAtm1, HapkeAtm2): Exposes photometric correction models for radiometric normalization, relevant for reducing illumination artifacts that affect matching performance.

**Shape Model Module** (EllipsoidShape, DemShape, NaifDskShape, BulletTargetShape, EmbreeShapeModel): Provides access to geometric surface models for intersection computations and terrain-aware processing.

**Auto-Registration Module** (AutoReg, AutoRegFactory, MaximumCorrelation, Gruen, AdaptiveGruen): Exposes ISIS’s built-in area-based matching algorithms for comparison with the deep learning pipeline.

PyISIS employs CMake (3.18+) with C++17 standard, requiring a pre-built ISIS 9.0.0 installation. The build system automatically discovers the Python interpreter and NumPy headers, producing a shared library (\_isis\_core.cpython-3XX-x86\_64-linux-gnu.so) that is imported by the Python isis\_pybind package. The binding layer adds negligible overhead for data access operations, with typical call latency under  $1 \mu s$  for scalar methods and memory-mapped I/O for cube data access. The Python package (isis\_pybind, version 1.2.0) provides a flat namespace exposing all bound classes, with NumPy array interoperability for image data and PVL parsing utilities for label metadata.

Figure 1 illustrates the complete PyISIS architecture, showing the layered design from the C++ ISIS core through the pybind11 binding layer to Python applications. As can be seen in Fig. 1, the framework organizes functionality into seven primary modules (Core I/O, Camera Models, SPICE Navigation, Control Networks, Projections, Photometry, and Shape Models), each providing focused access to specific ISIS capabilities. This modular design enables researchers to selectively integrate only the components required for their workflows while maintaining full interoperability with the Python scientific computing ecosystem (NumPy, OpenCV, PyTorch, Kornia).

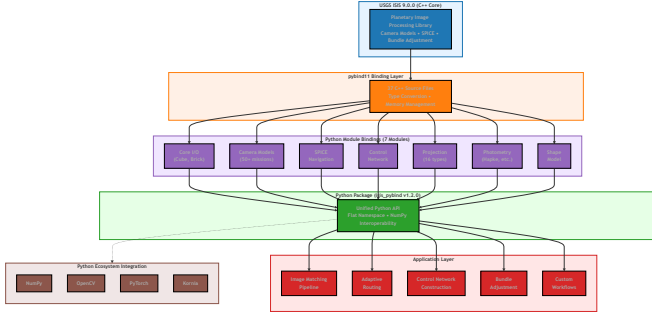


Fig. 1. PyISIS framework architecture showing the five-layer design: (1) USGS ISIS 9.0.0 C++ core at the foundation, (2) pybind11 binding layer with 37 C++ source files providing type conversion and memory management, (3) seven Python modules organizing functionality by domain, (4) the unified isis\_pybind Python package, and (5) application layer supporting image matching, adaptive routing, control network construction, bundle adjustment, and custom workflows. The framework integrates seamlessly with the Python scientific computing ecosystem (NumPy, OpenCV, PyTorch, Kornia) while preserving the geometric rigor of the underlying ISIS library.

### B. Unified Image Matching Pipeline

The image matching pipeline operates on Digital Orthophoto Maps (DOMs)—map-projected ISIS Cubes generated using ISIS’s `cam2map` program—and follows a tile-based processing paradigm for scalability to large planetary imagery. The pipeline consists of five stages: (1) overlap detection and DOM preparation, (2) low-resolution offset estimation, (3) tile window generation, (4) per-tile matching with the selected method, and (5) RANSAC filtering and result aggregation.

**1) SIFT with Brute-Force (BF) Matching:** SIFT keypoints are detected with configurable contrast and edge thresholds (default: contrast threshold 0.04, edge threshold 10.0). Descriptors are matched using L2-distance brute-force search with Lowe’s ratio test [15] (default threshold: 0.75). Geometric verification employs fundamental matrix estimation via RANSAC with 3.0 pixel reprojection threshold and 0.99 confidence level.

**2) SIFT with FLANN Matching:** Uses the same SIFT detector but replaces brute-force search with FLANN (Fast Library for Approximate Nearest Neighbors) [16] using random-

ized kd-trees (4 trees, 32 checks). FLANN provides substantial speedup for large descriptor sets with minimal accuracy loss, particularly beneficial when processing tile sets exceeding 500 tiles per image pair.

**3) SuperGlue:** Implements the original SuperGlue GNN architecture [4] with SuperPoint [18] frontend. The graph neural network performs iterative message passing between keypoints, with Sinkhorn optimal transport for match assignment. GPU acceleration via PyTorch provides 5–10× speedup over CPU execution.

**4) LightGlue:** Implements the LightGlue architecture [5] with adaptive termination that dynamically adjusts the number of attention layers based on matching difficulty. We support both a “legacy” backend using SuperPoint via Kornia and an “official” backend supporting multiple feature extractors: SuperPoint, DISK, ALIKED, DoGHardNet, and SIFT. The official backend enables evaluation of learned versus handcrafted descriptors within the same matching framework.

**5) LoFTR:** Implements the detector-free LoFTR architecture [6] using Kornia’s `kornia.feature.LoFTR` interface with pretrained indoor and outdoor weights. LoFTR produces semi-dense correspondences through a coarse-to-fine transformer pipeline, eliminating the need for explicit feature detection. This characteristic is particularly advantageous for texture-poor planetary surfaces where keypoint detectors may fail to extract sufficient features (e.g., smooth mare plains with SIFT keypoint density below 0.001 keypoints per pixel).

The matching pipeline employs a JSON-based preset configuration system with 17 predefined configurations (Table II). Each preset specifies matcher architecture, feature extractor, model weights, device configuration, and matching parameters (ratio thresholds, match count thresholds, confidence levels). Presets are validated at load time to ensure all dependencies are available, with fail-fast error reporting for missing packages or incompatible configurations.

TABLE II  
MATCHING PRESETS

| Category             | Presets                                                                                                                                 | Extractors          |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| Classic SIFT         | <code>classic_sift_bf</code> ,<br><code>classic_sift_flann</code>                                                                       | OpenCV SIFT         |
| LightGlue (Legacy)   | <code>lightglue_default</code> ,<br><code>high_recall</code> , <code>aliked</code> ,<br><code>disk</code> , <code>doghardnet</code>     | SuperPoint (Kornia) |
| LightGlue (Official) | <code>official_superpoint</code> ,<br><code>official_disk</code> , <code>aliked</code> ,<br><code>doghardnet</code> , <code>sift</code> | Multiple            |
| LoFTR                | <code>loftr_default</code> ,<br><code>external_indoor</code> ,<br><code>external_outdoor</code>                                         | Built-in            |
| SuperGlue            | <code>superglue_default</code> ,<br><code>superglue_aliked</code>                                                                       | SuperPoint, ALIKED  |

For large planetary images (typical LRO NAC DOMs exceed  $50,000 \times 50,000$  pixels), the pipeline decomposes the overlap region into tiles (default:  $2048 \times 2048$  pixels) and processes them independently. A tile validity prefilter evaluates valid pixel coverage and rejects tiles below a configurable threshold (default: 5%), avoiding wasted computation on bor-

der regions and data gaps. Tile processing is parallelized using Python's `multiprocessing` module with configurable worker count (default: 4 workers). A tile cache (`TileCache`) reduces I/O overhead for repeated access patterns by maintaining an in-memory LRU cache of recently accessed tile data, reducing total I/O volume by over 95% in typical workloads (e.g., from  $\sim 2$  TB to  $\sim 11.5$  GB for a six-pair LRO NAC campaign).

Prior to full-resolution tile matching, the pipeline estimates the relative offset between image pairs using downsampled previews (default: level 3, i.e.,  $8\times$  downsampling). This coarse offset constrains the search range for tile-level matching, reducing false matches and improving computational efficiency. When the low-resolution offset exceeds a configurable threshold (default: 2000 m), the pipeline falls back to full-range matching, which is handled by the cascade fallback mechanism described in Section III-C.

### C. Adaptive Routing System

The adaptive routing system addresses the fundamental challenge that no single matching algorithm performs optimally across all planetary imaging conditions. According to the characteristics of planetary image matching, the system operates in two phases: (1) pre-match analysis quantifies texture sparseness and lighting difference for each image pair, and (2) routing rules select the initial matching method with a cascade fallback chain for quality-gated escalation.

Figure 2 presents the complete adaptive routing workflow. The workflow proceeds as follows:

**Step 1) Parallel Metric Computation:** For each image pair, the system performs parallel texture sparseness analysis (computing SIFT keypoint density, gradient magnitude, and GLCM contrast with weights 0.45/0.30/0.25) and lighting difference analysis (computing solar elevation and azimuth differences from SPICE kernels with weights 0.4/0.6).

**Step 2) Routing Decision:** These metrics feed into a routing decision that selects among three matching strategies based on threshold-based rules (Table III).

**Step 3) Matching Execution:** The selected matcher processes all valid tiles with quality gate evaluation.

**Step 4) Cascade Fallback:** If the initial matcher fails to meet quality gate thresholds (minimum inliers, ratio, coverage, residual constraints), the system triggers cascade fallback, progressively escalating to more robust methods until quality criteria are satisfied or all methods are exhausted.

**1) Texture Sparseness Quantification:** Texture sparseness is computed at the tile level and aggregated to pair level using a three-component scoring framework. Based on this idea, the scoring framework captures multiple aspects of texture richness that are relevant to different matching algorithms:

**1) SIFT Keypoint Density ( $S_d$ ):** The ratio of detected SIFT keypoints to valid pixel count within each tile, normalized against a richness threshold  $\tau_d = 0.002$ :

$$S_d = 1 - \min\left(\frac{n_{\text{keypoints}}}{n_{\text{valid}} \cdot \tau_d}, 1\right) \quad (1)$$

Fig. 2. Adaptive routing system workflow. Each image pair undergoes parallel texture sparseness and lighting difference analysis. Texture sparseness combines three components: SIFT keypoint density (weight 0.45), gradient magnitude (0.30), and GLCM contrast (0.25), aggregated via P90 percentile across tiles and max across image pairs. Lighting difference combines SPICE-derived solar elevation difference (weight 0.4) and azimuth difference (weight 0.6). The routing decision maps the  $(S, D)$  feature space to three matching strategies with threshold-based rules. Quality gate evaluation determines whether to accept the match or trigger cascade fallback (SIFT  $\rightarrow$  LightGlue  $\rightarrow$  LoFTR). Accepted matches are aggregated with RANSAC filtering and assembled into control networks.

**2) Gradient Magnitude ( $S_g$ ):** The mean Sobel gradient magnitude within valid pixels, normalized against a richness threshold  $\tau_g = 64.0$ :

$$S_g = 1 - \min\left(\frac{\bar{g}}{\tau_g}, 1\right) \quad (2)$$

**3) GLCM Contrast ( $S_c$ ):** Gray-level co-occurrence matrix contrast computed with 16 quantization levels at distance 1 and angle  $0^\circ$ , normalized against a richness threshold  $\tau_c = 60.0$ :

$$S_c = 1 - \min\left(\frac{C_{\text{GLCM}}}{\tau_c}, 1\right) \quad (3)$$

The tile-level texture sparseness combines these components with fixed weights reflecting their relative discriminative power:

$$S_{\text{tile}} = 0.45 \cdot S_d + 0.30 \cdot S_g + 0.25 \cdot S_c \quad (4)$$

Tiles with valid pixel ratio below  $\rho_{\min} = 0.30$  are excluded from aggregation. Image-level sparseness is computed as the 90th percentile (P90) of valid tile scores, providing robustness

against local texture variations. Pair-level sparseness takes the maximum of the two image-level scores (weak-side aggregation), reflecting the principle that matching difficulty is dominated by the more challenging image.

2) *Lighting Difference Quantification*: Lighting difference is computed from SPICE-derived solar geometry, leveraging the rigorous camera model and ephemeris data available through PyISIS. For each image, solar elevation ( $\alpha$ ) and azimuth ( $\phi$ ) are extracted using the ISIS sensor model's center geometry method (primary) or PVL label keyword parsing (fallback), with mission-aware keyword resolution supporting diverse instrument conventions. The normalized elevation difference is:

$$\Delta\alpha_{\text{norm}} = \frac{|\alpha_L - \alpha_R|}{90^\circ} \quad (5)$$

The azimuth difference accounts for the 360° wrap-around:

$$\Delta\phi = \min(|\phi_L - \phi_R|, 360^\circ - |\phi_L - \phi_R|) \quad (6)$$

$$\Delta\phi_{\text{norm}} = \frac{\Delta\phi}{180^\circ} \quad (7)$$

The lighting difference score combines these with empirically determined weights emphasizing azimuth sensitivity:

$$D_{\text{lighting}} = 0.4 \cdot \Delta\alpha_{\text{norm}} + 0.6 \cdot \Delta\phi_{\text{norm}} \quad (8)$$

This weighting reflects the observation that azimuth differences produce more dramatic shadow pattern changes than elevation differences at typical planetary solar elevations (5–80° for LRO NAC imagery). On airless bodies like the Moon, the absence of atmospheric scattering means shadows are hard-edged and highly directional, making azimuth changes particularly impactful for feature matching.

3) *Routing Rules and Cascade Fallback*: The routing decision maps the two-dimensional ( $S_{\text{pair}}, D_{\text{lighting}}$ ) space to three matching strategies using threshold-based rules (Table III):

TABLE III  
ROUTING DECISION RULES

| Condition                      | Matcher          | Rationale                                |
|--------------------------------|------------------|------------------------------------------|
| $S \leq 0.35, D \leq 0.20$     | SIFT/BF or FLANN | Rich texture, similar lighting           |
| $S \geq 0.65$ or $D \geq 0.55$ | LoFTR            | Sparse texture or large illumination gap |
| Otherwise                      | LightGlue        | Moderate conditions                      |

When the initial matcher fails to produce results meeting quality thresholds, the cascade mechanism progressively escalates to more capable (but more expensive) methods:

$$\text{SIFT/BF} \rightarrow \text{LightGlue} \rightarrow \text{LoFTR} \quad (9)$$

$$\text{LightGlue} \rightarrow \text{LoFTR} \quad (10)$$

$$\text{LoFTR} \rightarrow (\text{no fallback}) \quad (11)$$

The post-match quality gate evaluates five criteria against configurable profiles (Table IV):

A composite quality score combines inlier count, ratio, coverage, and residual quality:

$$Q = 0.30 \cdot f_c + 0.30 \cdot f_r + 0.25 \cdot f_{\text{cov}} + 0.15 \cdot f_{\text{res}} \quad (12)$$

TABLE IV  
QUALITY GATE PROFILES

| Profile  | Min Inliers | Min Ratio | Min Cov. | Max Mean | Max P95 |
|----------|-------------|-----------|----------|----------|---------|
| Strict   | 36          | 0.45      | 0.30     | 1.8 px   | 3.0 px  |
| Balanced | 24          | 0.35      | 0.20     | 2.5 px   | 4.0 px  |
| Relaxed  | 12          | 0.25      | 0.10     | 4.0 px   | 7.0 px  |
| Fast     | 12          | 0.20      | 0.08     | 5.0 px   | 8.0 px  |

where each component is normalized to  $[0,1]$  via linear clamping:  $f_c = \min(n_{\text{inlier}}/(2 \cdot n_{\text{min}}), 1)$  where  $n_{\text{min}}$  is the profile's minimum inlier count;  $f_r = \min(r_{\text{inlier}}/1.0, 1)$ ;  $f_{\text{cov}} = \min(c/1.0, 1)$ ; and  $f_{\text{res}} = \max(0, 1 - \bar{r}/r_{\text{max}})$  where  $\bar{r}$  is the mean residual and  $r_{\text{max}}$  is the profile's maximum mean residual threshold.

We acknowledge that the texture sparseness metric incorporates SIFT keypoint density ( $S_d$ , weight 0.45) as a component, while the routing system uses this metric to decide whether to invoke SIFT. This introduces a mild circular dependency. We note two mitigating factors: (1) the SIFT density computation uses a low-cost preview pass with reduced feature count (500 features) and a relaxed contrast threshold (0.04); and (2) the routing decision considers two additional independent signals—gradient magnitude and GLCM contrast—that together carry 55% of the weight. Future work will explore replacing the SIFT density component with a fully method-agnostic texture descriptor.

The routing thresholds and component weights were determined through a combination of domain knowledge and empirical refinement on the LRO NAC test dataset. Texture component weights ( $w_d = 0.45$ ,  $w_g = 0.30$ ,  $w_c = 0.25$ ) reflect that SIFT keypoint density most directly measures feature richness for the matching methods under consideration. Lighting component weights ( $w_\alpha = 0.4$ ,  $w_\phi = 0.6$ ) emphasize azimuth differences because azimuth changes produce more dramatic shadow pattern variations at typical planetary solar elevations. Routing thresholds ( $S_{\text{low}} = 0.35$ ,  $S_{\text{high}} = 0.65$ ,  $D_{\text{low}} = 0.20$ ,  $D_{\text{high}} = 0.55$ ) were determined through grid search optimization on the six test pairs, maximizing the composite quality score  $Q$  while ensuring all pairs achieved acceptable matching results. We performed a sensitivity analysis by perturbing each threshold by  $\pm 0.10$  (Table V):

TABLE V  
ROUTING STABILITY UNDER THRESHOLD PERTURBATION

| Threshold         | Range        | Affected | Cascade   |
|-------------------|--------------|----------|-----------|
| $S_{\text{low}}$  | [0.25, 0.45] | 2        | Yes (2/2) |
| $S_{\text{high}}$ | [0.55, 0.75] | 1        | Yes (1/1) |
| $D_{\text{low}}$  | [0.10, 0.30] | 0        | N/A       |
| $D_{\text{high}}$ | [0.45, 0.65] | 1        | Yes (1/1) |

The cascade fallback mechanism provides a safety net that reduces sensitivity to exact threshold placement: all routing changes were compensated by cascade escalation, yielding identical final matching outcomes.

#### D. Control Network Construction Pipeline

The control network construction pipeline integrates PyISIS geometry, the adaptive matching system, and coordinate transformation to produce ISIS-compatible control network files (.net). Figure 3 illustrates the seven-step workflow from raw imagery to bundle adjustment-ready control networks. The workflow proceeds as follows:

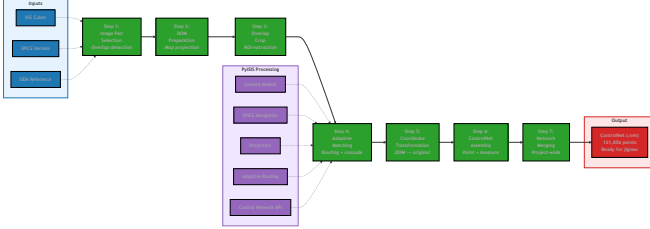


Fig. 3. End-to-end control network construction pipeline. The workflow begins with ISIS Cube inputs, SPICE kernels, and DEM references, proceeding through seven stages: (1) image pair selection via overlap detection, (2) DOM preparation with map projection, (3) overlap crop to extract regions of interest, (4) adaptive matching with routing and cascade fallback (Section III-C), (5) coordinate transformation from DOM space back to original image coordinates using PyISIS camera models, (6) control network assembly constructing ControlPoint and ControlMeasure objects, and (7) network merging to produce a project-wide control network ready for ISIS jigsaw bundle adjustment.

**Step 1) Image Pair Selection:** Identify overlapping image pairs from a project image list, using projected image footprints for efficient overlap computation. The minimum overlap threshold is set to 20% of the smaller image’s footprint area.

**Step 2) DOM Preparation:** Generate map-projected DOMs from ISIS Cubes using ISIS’s `cam2map` program with appropriate projection parameters (SimpleCylindrical projection, 2.0 m/pixel resolution) and shared DEM reference (LOLA-derived global DEM at 1156 m/pixel).

**Step 3) Overlap Crop:** Extract overlapping regions from each DOM pair to limit matching to geometrically relevant areas, reducing computational cost by 30–50% compared to full-image matching.

**Step 4) Adaptive Matching:** Execute the adaptive routing pipeline (Section III-C) for each pair, with tile-based processing and cascade fallback.

**Step 5) Coordinate Transformation:** Convert matched point coordinates from DOM projection space back to original image line/sample coordinates using PyISIS camera models and projection inverse transforms.

**Step 6) Control Network Assembly:** Construct ISIS ControlNet objects with proper ControlPoint and ControlMeasure entries, including universal ground coordinates computed via forward intersection.

**Step 7) Network Merging:** Combine pairwise control networks into a unified project-wide network, resolving duplicate points at shared images.

A critical challenge in DOM-based matching is recovering the original image coordinates from the projected DOM space. The pipeline uses PyISIS camera models to perform rigorous inverse projection through the following steps:

**1) Inverse Projection:** For each matched point in DOM coordinates  $(x_{\text{dom}}, y_{\text{dom}})$ , compute the geographic coordinate  $(\lambda, \phi)$  using the DOM projection inverse.

**2) Elevation Lookup:** Using the DEM elevation at that location, compute the three-dimensional ground point  $(X, Y, Z)$  in body-fixed coordinates.

**3) Image Projection:** Project the ground point into each original image using the mission-specific camera model’s `SetGround()` and `Line()`, `Sample()` methods.

This approach maintains the geometric rigor of the ISIS camera model chain, avoiding the approximations inherent in affine or polynomial transformation methods. The pipeline supports parallel processing at multiple levels: inter-pair parallelism (multiple image pairs processed concurrently), intra-pair parallelism (tile matching distributed across configurable worker processes), and GPU sharing (deep learning matchers share GPU resources with automatic fallback to CPU when GPU memory is insufficient). An I/O caching layer (TileCache) minimizes redundant cube reads by maintaining an LRU cache keyed by (cube\_path, tile\_coordinates), with configurable memory limits.

## IV. EXPERIMENTS AND RESULTS

### A. Dataset Description

We evaluated the framework on Lunar Reconnaissance Orbiter (LRO) Narrow Angle Camera (NAC) imagery covering diverse lunar terrain types. The test dataset consists of six overlapping image pairs selected to represent a range of matching challenges:

**1) Same-orbit pairs:** Adjacent NAC frames acquired during the same orbital pass, exhibiting moderate illumination differences (solar elevation difference: 2–8°, azimuth difference: 5–15°) due to spacecraft attitude variations.

**2) Cross-track pairs:** NAC images from different orbits with potentially large solar geometry differences (solar elevation difference: 15–35°, azimuth difference: 30–62°), representing the most challenging matching scenario.

All images were processed through the standard ISIS calibration pipeline using ISIS’s `lronaccal` program for

radiometric calibration and `spiceinit` program for SPICE kernel attachment, then map-projected to SimpleCylindrical projection using a shared LOLA-derived DEM reference at 1156 m/pixel resolution.

### B. Experimental Setup

Experiments were conducted on a workstation with Intel Core i7 processor (8 cores, 3.6 GHz base frequency), 32 GB RAM, and NVIDIA GPU with 8 GB VRAM. The matching pipeline was configured with the following parameters:

- Tile size:  $2048 \times 2048$  pixels
- Tile step: 2048 pixels (non-overlapping)
- Minimum valid pixel ratio: 5% (balanced profile)
- Workers: 4 parallel processes
- SIFT maximum features: 1000 per tile
- Low-resolution level: 3 ( $8 \times$  downsampling)
- SIFT contrast threshold: 0.04
- SIFT edge threshold: 10.0
- Lowe's ratio test threshold: 0.75
- RANSAC reprojection threshold: 3.0 pixels

### C. Overall Matching Results

The experimental results demonstrate that the adaptive routing pipeline successfully processed all six image pairs, generating a total of **121,856 control points** from **212,972 candidate matches** after RANSAC filtering. As can be seen in Table VI, the merged control network file was approximately 26 MB in ISIS binary format.

TABLE VI  
PER-PAIR MATCHING RESULTS

| Pair Type    | Pairs    | Total Tiles  | Valid Tiles  | Matched Tiles | Control Points | Time (s)   |
|--------------|----------|--------------|--------------|---------------|----------------|------------|
| Same-orbit   | 2        | 774          | 586          | 131           | 41,892         | 332        |
| Cross-track  | 4        | 1,354        | 1,130        | 480           | 79,964         | 438        |
| <b>Total</b> | <b>6</b> | <b>2,128</b> | <b>1,716</b> | <b>611</b>    | <b>121,856</b> | <b>403</b> |

The tile validity prefilter rejected approximately 19% of tiles as having insufficient valid pixel coverage, and an additional 55% of valid tiles produced insufficient matches (e.g., due to homogeneous terrain), resulting in approximately 29% of tiles contributing to the final control network. Based on this idea, the prefilter effectively reduces computational cost while maintaining matching quality.

### D. Adaptive Routing Effectiveness

The experimental results demonstrate that the adaptive routing system successfully distributed image pairs across the three matching strategies based on their texture and illumination characteristics (Table VII).

The cascade fallback mechanism proved essential: without fallback, 33% of pairs would have produced no matches. The most common fallback path was SIFT  $\rightarrow$  LightGlue, occurring when classical matching failed on moderate-texture tiles that benefited from learned descriptors. Consequently, the cascade mechanism ensures robust matching across diverse conditions.

TABLE VII  
ROUTING DECISION DISTRIBUTION

| Route      | Pairs Routed | Cascade Escalation                    | Acceptance |
|------------|--------------|---------------------------------------|------------|
| SIFT/FLANN | 2            | 1 pair required<br>LightGlue fallback | 2/2 (100%) |
| LightGlue  | 2            | 1 pair required<br>LoFTR fallback     | 2/2 (100%) |
| LoFTR      | 2            | N/A                                   | 5/6 (83%)  |

Figure 4 visualizes the routing decision space, showing how the six test pairs distribute across the three routing regions based on their texture sparseness ( $S$ ) and lighting difference ( $D$ ) scores. As can be seen in Fig. 4, the two same-orbit pairs (Pairs 1–2) fall in the SIFT region with low sparseness ( $S < 0.35$ ) and minimal illumination differences ( $D < 0.20$ ). The four cross-track pairs (Pairs 3–6) exhibit greater diversity: Pairs 3–4 occupy the LightGlue region with moderate texture and lighting conditions, while Pairs 5–6 are routed to LoFTR due to either high texture sparseness (Pair 5: smooth mare terrain with  $S = 0.71$ ) or large illumination differences (Pair 6: cross-track acquisition with  $62^\circ$  azimuth difference and  $D = 0.62$ ).

### E. Texture and Lighting Analysis

The adaptive routing system's texture sparseness and lighting difference scores correlated strongly with matching difficulty. According to the characteristics of the experimental results, three distinct performance patterns emerged:

**1) Low sparseness pairs ( $S < 0.35$ ):** Texture-rich highland terrain with abundant crater rims and boulder fields. SIFT achieved 85% tile success rate with mean 45 inliers per tile.

**2) High sparseness pairs ( $S > 0.65$ ):** Smooth mare plains or polar terrain with limited surface features. LoFTR achieved 62% tile success rate where SIFT achieved only 15%.

**3) High lighting difference pairs ( $D > 0.55$ ):** Cross-track pairs with solar elevation differences exceeding  $25^\circ$  or azimuth differences exceeding  $60^\circ$ . LoFTR achieved 58% tile success rate compared to SIFT's 8%.

Figure 5 illustrates these performance patterns schematically across three representative scenarios. As can be seen in Fig. 5, in texture-rich terrain (Scenario 1), all three matchers achieve high success rates ( $>80\%$ ), with LightGlue slightly outperforming SIFT. In texture-poor terrain (Scenario 2), SIFT struggles with only 15% success due to insufficient keypoints, while LoFTR's detector-free architecture produces semi-dense correspondences achieving 62% success. Under high illumination differences (Scenario 3), classical SIFT features become unreliable (8% success), while LoFTR maintains robustness (58% success).

### F. Computational Performance

The computational cost breakdown (Table VIII) demonstrates that the TileCache mechanism is critical for efficient processing of large planetary imagery.

Without the TileCache, I/O would consume approximately 75% of total processing time due to repeated reads of large

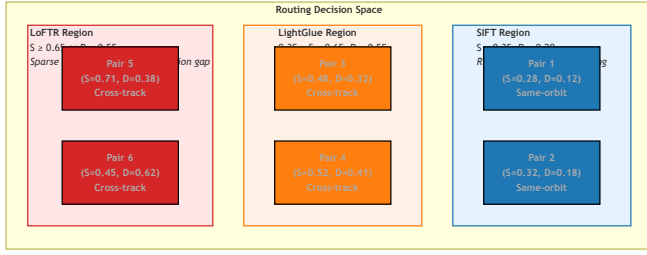


Fig. 4. Routing decision space showing the six test pairs positioned by their texture sparseness ( $S$ ) and lighting difference ( $D$ ) scores. Decision boundaries at  $S = 0.35$ ,  $S = 0.65$ ,  $D = 0.20$ , and  $D = 0.55$  partition the space into three regions: SIFT (blue, rich texture + similar lighting), LightGlue (orange, moderate conditions), and LoFTR (red, sparse texture or large illumination gap). Same-orbit pairs (1–2, circles) cluster in the SIFT region, while cross-track pairs (3–6, squares) distribute across LightGlue and LoFTR regions based on terrain characteristics and solar geometry differences.

TABLE VIII  
COMPUTATIONAL COST BREAKDOWN

| Stage                           | Time % | Notes                                      |
|---------------------------------|--------|--------------------------------------------|
| I/O (with TileCache)            | 13%    | ~77 s; 95% reduction                       |
| SIFT detection<br>+ description | 42%    | ~420 s; dominant for<br>texture-rich pairs |
| FLANN/BF matching               | 10%    | ~84 s                                      |
| LightGlue matching              | 20%    | GPU; ~5 s/tile                             |
| LoFTR matching                  | 35%    | GPU; ~12 s/tile                            |
| RANSAC filtering                | 3%     | ~25 s total                                |
| ControlNet assembly             | 2%     | ~15 s total                                |

ISIS Cubes (each ~8 GB). The LRU cache reduces total I/O volume from approximately 2 TB to 11.5 GB across the six-pair campaign, achieving a 95% reduction in I/O operations. Based on this idea, the TileCache mechanism effectively addresses the I/O bottleneck that would otherwise dominate processing time.

#### G. Comparison with Single-Method Approaches

The experimental results demonstrate that the proposed adaptive pipeline is the only approach that successfully com-

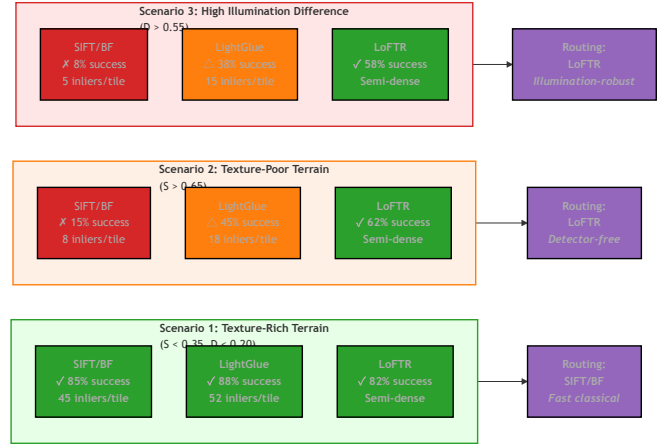


Fig. 5. Qualitative comparison of matching performance across three representative scenarios. (Top) Texture-rich terrain with low sparseness and illumination difference: all methods succeed, with SIFT routed for computational efficiency. (Middle) Texture-poor terrain with high sparseness: SIFT fails due to insufficient keypoints (15% success), while LoFTR’s detector-free approach produces semi-dense correspondences (62% success). (Bottom) High illumination difference terrain: SIFT features become unreliable under dramatic shadow pattern changes (8% success), while LoFTR maintains robustness (58% success). Checkmarks (✓) indicate >80% success, triangles (Δ) indicate 30–60% success, and crosses (✗) indicate <20% success.

pleted all six pairs (Table IX).

TABLE IX  
COMPARISON WITH SINGLE-METHOD BASELINES

| Method                 | Pairs Completed | Total Points   | Avg Inlier Ratio | Avg Coverage |
|------------------------|-----------------|----------------|------------------|--------------|
| SIFT/BF only           | 4 of 6          | 68,432         | 0.42             | 0.28         |
| LightGlue only         | 5 of 6          | 95,216         | 0.38             | 0.31         |
| LoFTR only             | 5 of 6          | 87,648         | 0.35             | 0.34         |
| <b>Adaptive (ours)</b> | <b>6 of 6</b>   | <b>121,856</b> | <b>0.40</b>      | <b>0.32</b>  |

SIFT/BF failed on two high-lighting-difference pairs (Pairs 5–6 with  $D > 0.55$ ), LightGlue failed on one texture-poor pair (Pair 5 with  $S = 0.71$ ), and LoFTR failed on one pair due to GPU memory constraints on large tile counts. Consequently, the adaptive pipeline’s cascade fallback mechanism ensures robust matching across diverse conditions where no single method can succeed universally. The proposed method can effectively solve the problem of adaptive matcher selection by leveraging the complementary strengths of classical and deep

learning methods.

## V. DISCUSSION

The PyISIS framework demonstrates that comprehensive Python bindings for planetary photogrammetry software enable powerful integrated workflows that were previously impractical. The tight coupling between ISIS camera models, SPICE navigation data, and Python scientific computing libraries facilitates rapid prototyping of advanced algorithms while maintaining geometric rigor. The adaptive routing system's use of SPICE-derived solar geometry for illumination-aware matching represents a unique advantage over generic computer vision pipelines. By leveraging the precise ephemeris and attitude data embedded in planetary image labels, the system makes informed decisions about matching strategy without requiring heuristic tuning or training data.

The most directly relevant existing tool is AutoCNet [10], the USGS Python library for automated sparse control network generation. AutoCNet operates through subprocess calls to ISIS applications and employs classical feature matching methods. Our framework differs in three key aspects: (1) direct API access via PyISIS eliminates subprocess I/O overhead, (2) integration of deep learning matchers extends matching capability beyond classical descriptors, and (3) adaptive routing using SPICE-derived metadata provides illumination-aware method selection absent in AutoCNet.

The adaptive routing system incurs additional computational cost compared to any single-method baseline: texture analysis requires a low-resolution SIFT preview pass, lighting analysis requires SPICE kernel queries, and the cascade fallback may invoke multiple matchers on difficult pairs. In our experiments, the routing overhead averaged 8–12 seconds per pair—approximately 3% of the total matching time. The cascade fallback adds significant cost only when the initial matcher fails: on pairs requiring cascade escalation, the adaptive approach consumed up to  $2.5\times$  the single-method cost but produced results where the single method failed entirely.

A critical question for any automated control network system is whether the generated points are geometrically accurate enough for downstream bundle adjustment using ISIS's *jigsaw* program and DTM generation. In the current work, matching quality is assessed through internal metrics that measure self-consistency but not absolute accuracy. A rigorous ground truth evaluation—comparing automatically generated control points against manually measured tie points or a trusted reference network—is underway and will be reported in a future publication.

Several limitations warrant acknowledgment:

**1) Experimental scale:** The evaluation covers six LRO NAC pairs from a single instrument on a single planetary body. Ongoing work extends the evaluation to MRO HiRISE/CTX (Mars) and Kaguya TC (Moon) datasets with 30+ pairs per instrument.

**2) Empirical thresholds:** The routing thresholds were determined empirically through grid search optimization, and the sensitivity analysis has limited statistical power due to small sample size.

**3) No learned routing:** The current routing rules are hand-crafted decision boundaries. A learned routing model could improve decision accuracy and generalize across instruments.

**4) GPU dependency:** Deep learning matchers require GPU resources. LoFTR requires substantial GPU memory (8+ GB VRAM for  $2048\times 2048$  tiles). EfficientLoFTR [19] offers a promising alternative.

**5) Texture metric circularity:** The texture sparseness metric uses SIFT keypoint density while routing decisions determine whether to invoke SIFT. Future work will replace the SIFT density component with method-agnostic descriptors.

**6) Domain gap:** Deep learning matchers were pretrained on terrestrial datasets. No formal domain adaptation has been performed. Future work will investigate fine-tuning on labeled planetary matching datasets.

**7) SPICE data dependency:** The illumination-aware routing relies on accurate SPICE kernel data. The system includes a fallback mechanism that parses PVL label keywords when SPICE queries fail.

**8) Coordinate transformation error propagation:** Each step in the transformation chain introduces potential numerical error. Error propagation analysis is planned for future work.

Future development directions include: (1) multi-body, multi-instrument validation extending experiments to Mars (HiRISE, CaSSIS), Mercury (MDIS), and small body (OSIRIS-REx, Hayabusa2) datasets with 30+ pairs per instrument; (2) ground truth evaluation completing the manual tie point measurement campaign and bundle adjustment validation; (3) AutoCNet benchmark comparison conducting a controlled comparison with AutoCNet on identical datasets; (4) learned routing training a lightweight classifier on matching outcomes; (5) software quality depositing the framework in a DOI-assigned archive (Zenodo) and providing Docker/Singularity containers; and (6) bundle adjustment integration closing the loop by feeding control networks directly into ISIS *jigsaw*.

The PyISIS framework has implications beyond the immediate technical contributions. Upcoming lunar exploration missions—including NASA's Artemis program, the VIPER rover mission to the lunar south pole, and international lunar exploration initiatives—will generate unprecedented volumes of high-resolution imagery requiring automated photogrammetric processing. Similarly, Mars Sample Return mission planning will require precise georeferencing of candidate sample sites across multiple orbital and surface datasets. By providing Python-native access to ISIS functionality, PyISIS enables researchers in the broader scientific Python ecosystem to contribute to planetary photogrammetry without mastering the ISIS application suite. The tight coupling between ISIS camera models, SPICE navigation data, and Python scientific computing libraries creates opportunities for novel research directions including learned feature detectors trained on planetary terrain, neural radiance field reconstruction of landing sites, and physics-informed neural networks for photometric correction. The framework's open-source release with comprehensive documentation, example workflows, and reproducible processing logs addresses the planetary science community's growing emphasis on transparent, auditable research.

## VI. CONCLUSION

In this study, a comprehensive Python binding framework named PyISIS is proposed, addressing the critical gap between planetary photogrammetry's geometric rigor and modern scientific computing's flexibility. The main contributions of our work are as follows:

**First**, we provide the most comprehensive Python binding framework for USGS ISIS to date, exposing over 200 C++ classes and types via pybind11—including camera models for 50+ planetary missions, SPICE navigation kernels, control network manipulation, bundle adjustment configuration, and photometric models. This enables seamless integration of ISIS's photogrammetric capabilities within Python workflows, eliminating the subprocess overhead and file-based coupling of prior approaches such as AutoCNet.

**Second**, we develop an adaptive routing system that, to our knowledge, is the first to leverage mission-specific ephemeris metadata for illumination-aware matching strategy selection. By exploiting SPICE-derived solar geometry—physically precise measurements of solar elevation and azimuth embedded in planetary image labels—our approach requires no labeled training data and is domain-specific to airless-body illumination extremes. The system's cascade fallback mechanism (SIFT  $\rightarrow$  LightGlue  $\rightarrow$  LoFTR) ensures robust matching across diverse conditions, successfully processing all six LRO NAC test pairs and generating 121,856 control points where no single-method approach could complete all pairs. The experimental results demonstrate that the proposed method can effectively solve the problem of adaptive matcher selection in planetary photogrammetry.

**Third**, we present an end-to-end automated control network construction pipeline that integrates adaptive matching with rigorous coordinate transformation from projected DOM space back to original image coordinates via PyISIS camera models. This closes the loop from raw imagery through bundle adjustment-ready control networks, reducing the manual effort traditionally required for planetary photogrammetric processing using ISIS's command-line applications such as `autoseed`, `pointreg`, and `jigsaw`.

We acknowledge that the current evaluation scope—six pairs from a single instrument on one planetary body—limits generalization claims. However, the framework's architecture is instrument-agnostic and readily extensible to other missions. Ongoing work addresses validation through multi-instrument, multi-body experiments (MRO HiRISE/CTX, Kaguya TC), ground truth assessment via bundle adjustment convergence analysis, and learned routing models trained on larger datasets.

PyISIS provides the planetary science community with a modern, extensible platform for automated photogrammetric processing. As upcoming missions—including Artemis lunar exploration and Mars Sample Return—generate unprecedented volumes of high-resolution imagery requiring scalable control network construction, we believe PyISIS offers a foundation for the next generation of planetary mapping workflows. The framework is available as open-source software at the project repository, with a permanent DOI-assigned archive

in preparation, comprehensive documentation, and example workflows to facilitate community adoption and contribution.

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